# ENVIRONMENTAL STUDIES

II Semester

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ASHWITH FRANK
ALPHAOMEGA STUDIO

## 2 Marks Questions:

**1. Acid Rain.**

Acid rain is rainfall that contains harmful acidic substances such as sulphur dioxide ($SO_2$) and nitrogen oxides ($NO_x$) mixed with water vapor in the atmosphere. It is mainly caused by pollution from industries, vehicles, and burning of fossil fuels.

**Effects of Acid Rain:**

- Damages plants and forests
- Pollutes water bodies
- Harms buildings and monuments
- Affects human health and animals

**Prevention:**

- Reduce air pollution
- Use clean energy sources
- Control industrial emissions

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**2. Biodiversity.**

Biodiversity refers to the variety of living organisms present on Earth, including plants, animals, and microorganisms. It also includes the diversity of ecosystems and genes.

**Importance:**

- Maintains ecological balance
- Provides food, medicine, and resources
- Supports environmental stability

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**3. Endangered Species.**

Endangered species are plants or animals that are at risk of becoming extinct due to habitat loss, hunting, pollution, and climate change.

**Examples:**

- Tiger
- Rhino

- Elephant

**Protection Methods:**

- Wildlife conservation
- Ban on hunting
- Protection of forests

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**4. Greenhouse Effect.**

The greenhouse effect is the process by which gases like carbon dioxide, methane, and water vapor trap heat in the Earth's atmosphere, making the planet warm.

**Causes:**

- Burning fossil fuels
- Deforestation
- Industrial pollution

**Effects:**

- Global warming
- Climate change
- Melting of glaciers and rising sea levels

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**5. Rainwater Harvesting**

Rainwater harvesting is the process of collecting and storing rainwater for future use instead of allowing it to flow away.

**Importance:**

- Increases groundwater level
- Saves water for daily use
- Reduces water scarcity
- Helps in agriculture

**Methods:**

- Rooftop rainwater collection
- Storage tanks

- Check dams

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**6. Natural Resources.**

Natural resources are materials and substances that occur naturally in the environment and are useful to humans.

**Examples:**

- Water
- Air
- Soil
- Forests
- Minerals

**Importance:**

- Support human life
- Provide energy and raw materials
- Maintain ecological balance

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**7. Renewable Resources:**

Renewable resources are natural resources that can be replaced or regenerated naturally within a short period of time.

**Examples:**

- Solar energy
- Wind energy
- Water
- Forests

**Importance:**

- Eco-friendly
- Reduce pollution
- Sustainable for future generations

## 10 Marks Questions:

**1. Meaning of Solid Waste Management.**

Solid Waste Management (SWM) is the process of collecting, transporting, treating, recycling, and disposing of waste materials produced by homes, industries, hospitals, shops, and other places in a safe and eco-friendly manner.

**The main aim of solid waste management is to:**

- Keep the environment clean
- Protect human health
- Reduce pollution
- Reuse valuable materials
- Properly dispose of harmful waste

**Examples of solid waste include:**

- Food waste
- Plastic bottles
- Paper
- Glass
- Metal cans
- Medical waste
- Electronic waste

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**2. Types of Solid Waste.**

**Municipal Waste:** Waste produced from houses, schools, offices, markets, and public places.

**Examples:**

- Food leftovers
- Paper
- Plastic covers
- Clothes

**Industrial Waste:** Waste generated from industries and factories.

**Examples:**

- Chemicals
- Metal scraps
- Ash
- Packaging materials

**Solid Waste Management:** Waste produced from farming and agricultural activities.

**Examples:**

- Crop residues
- Animal waste
- Pesticide containers

**Biomedical Waste:** Waste produced from hospitals, clinics, and laboratories.

**Examples:**

- Syringes
- Medicines
- Bandages
- Surgical gloves

**Electronic Waste (E-Waste):** Discarded electronic devices and accessories.

**Examples:**

- Old phones
- Computers
- Chargers
- Batteries

**Hazardous Waste:** Waste that is dangerous to humans and the environment.

**Examples:**

- Paints
- Chemicals
- Toxic materials
- Radioactive substances

**3. Methods of Solid Waste Management.**

**Recycling:** Converting waste materials into new useful products.

**Examples:**

- Recycling paper into notebooks
- Melting plastic to make containers

**Reuse:** Using items again instead of throwing them away.

**Examples:**

- Reusing bottles and bags
- Donating old clothes

**Composting:** Converting biodegradable waste into natural fertilizer.

**Examples:**

- Vegetable peels
- Dry leaves
- Food waste

**Incineration:** Burning waste at high temperatures to reduce its volume.

Used mainly for:

- Medical waste
- Hazardous waste

**Landfilling:** Dumping waste into specially designed landfill sites.

This method is used for waste that cannot be recycled or reused.

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**4. Importance of Solid Waste Management.**

**Protects Environment:** Proper waste disposal reduces land, air, and water pollution.

**Prevents Diseases:** Waste management helps control insects, mosquitoes, and bacteria that spread diseases.

**Saves Natural Resources:** Recycling and reusing materials reduce the need for raw resources.

**Reduces Pollution:** Scientific waste treatment minimizes harmful gases and toxic substances.

**Creates Employment:** Waste management industries provide jobs in:

- Recycling
- Collection
- Processing
- Manufacturing

**Improves Public Health and Cleanliness:** Clean surroundings create a healthier and better living environment.

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**5. Note on Plastic Pollution.**

**Introduction:** Plastic pollution refers to the accumulation of plastic materials in the environment that causes harm to humans, animals, and nature. Plastics are non-biodegradable, meaning they do not decompose easily and can remain in the environment for hundreds of years. Today, plastic pollution has become one of the major environmental problems across the world due to the excessive use of plastic products such as bags, bottles, wrappers, and containers.

**Causes of Plastic Pollution:**

- **Excessive Use of Plastic:** People use large amounts of single-use plastics in daily life.
- **Examples:**
  - Plastic bags
  - Water bottles
  - Food packaging
  - Straws
- **Improper Disposal of Waste:** Throwing plastic waste on roads, rivers, and open areas increases pollution.
- **Lack of Recycling:** Many plastic materials are not properly recycled and end up in landfills or oceans.
- **Industrial Waste:** Factories and industries release plastic waste into the environment.

**Effects of Plastic Pollution:**

- **Harm to Animals:** Animals and marine creatures often swallow plastic, mistaking it for food.
- **Examples:**

- Fish
- Sea turtles
- Birds
- Cows

- **Water Pollution:** Plastic waste blocks drains, rivers, and oceans, making water polluted and unsafe.
- **Soil Pollution:** Plastic reduces soil fertility and affects plant growth.
- **Air Pollution:** Burning plastic releases harmful gases that can cause breathing problems and diseases.
- **Threat to Human Health:** Tiny plastic particles called microplastics can enter food and drinking water, affecting human health.

**Prevention and Control of Plastic Pollution:**

- **Reduce Plastic Usage:** Avoid unnecessary use of plastic products.
- **Examples:**
  - Carry cloth bags
  - Use steel or glass bottles
- **Reuse Plastic Items:** Reuse containers and bags whenever possible.
- **Recycling:** Separate plastic waste and send it for recycling.
- **Public Awareness:** Educate people about the harmful effects of plastic pollution.
- **Government Rules:** Governments can ban single-use plastics and encourage eco-friendly alternatives.

**Importance of Controlling Plastic Pollution:**

- Protects the environment
- Saves animals and marine life
- Reduces pollution
- Improves public health
- Keeps surroundings clean and beautiful

**Conclusion:** Plastic pollution is a serious environmental issue that affects land, water, air, animals, and humans. Proper waste management, recycling, reducing plastic use, and spreading awareness are important steps to control this problem. Every individual should take responsibility to reduce plastic waste and protect the environment for future generations.

**6. Note on Water Shed Management.**

**Introduction:** Watershed Management is the process of proper use, conservation, and management of water, soil, and natural resources within a watershed area. A watershed is an area of land where all rainwater and streams drain into a common river, lake, or reservoir. The main aim of watershed management is to conserve water, prevent soil erosion, improve agriculture, and maintain ecological balance.

**Objectives of Watershed Management:**

- To conserve rainwater
- To improve groundwater levels
- To reduce soil erosion
- To increase agricultural production
- To protect natural resources
- To maintain environmental balance

**Components of Watershed Management:**

- **Soil Conservation:** Methods used to prevent soil erosion and maintain soil fertility.
- **Examples:**
    - Terracing
    - Contour ploughing
    - Afforestation
- **Water Conservation:** Storing and managing rainwater effectively.
- **Examples:**
    - Check dams
    - Rainwater harvesting
    - Farm ponds
- **Afforestation:** Planting trees to protect soil and improve water retention.
- **Sustainable Agriculture:** Using eco-friendly farming methods to increase productivity without damaging the environment.

**Importance of Watershed Management:**

- **Prevents Soil Erosion:** It reduces the loss of fertile topsoil caused by wind and water.

- **Conserves Water:** Helps in storing rainwater and improving groundwater recharge.
- **Supports Agriculture:** Provides sufficient water for irrigation and increases crop production.
- **Controls Floods and Droughts:** Proper management reduces flood damage and helps during dry seasons.
- **Protects Environment:** Maintains ecological balance and supports plant and animal life.
- **Improves Rural Development:** Creates employment opportunities and improves the living conditions of farmers.

**Methods of Watershed Management:**

- **Rainwater Harvesting:** Collecting and storing rainwater for future use.
- **Construction of Check Dams:** Small barriers built across streams to slow water flow and increase groundwater recharge.
- **Plantation Programs:** Growing trees and plants to conserve soil and water.
- **Controlled Grazing:** Limiting animal grazing to protect vegetation and soil.

**Advantages of Watershed Management:**

- Increases water availability
- Improves soil fertility
- Enhances agricultural yield
- Reduces environmental degradation
- Improves biodiversity

**Conclusion:** Watershed Management is essential for conserving water, protecting soil, and ensuring sustainable development. Proper management of natural resources helps improve agriculture, reduce environmental problems, and support human life. Everyone should participate in protecting and managing water resources for a better and greener future.